Sahaj Bhadja

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Summary

Software engineer experienced in **Python, C++, computer vision, autonomous-driving data pipelines, cloud storage, testable modular code, logging, and technical documentation**, delivering reliable tools for data quality and operator-facing workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built Python pipelines processing 5,000+ monthly jobs with structured logging, state tracking, validation, and cloud-storage uploads, increasing concurrent throughput 3x while reliably sustaining 99%+ reliability in production.
- Maintained modular services with reconciliation checks and documented data-quality workflows for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through indexing and query optimization.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed testable Python modules for vision, OCR, and file-ingestion workflows, processing 1,000+ images and documents per batch with automated tests, diagnostics, and reliably fault-tolerant execution at scale.
- Investigated model, timeout, and API failures across 10,000+ daily requests, implementing fallback paths and operator-facing logs while reliably reducing feedback turnaround time 60% for production users at scale.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Built Python and PyTorch pipelines processing 100,000+ autonomous-driving trajectory records, reliably improving long-horizon planning 18% through feature preparation, simulation, debugging, and iterative reproducible testing at production scale.
- Aligned ego, road, and nearby-agent signals to a shared coordinate frame, improving closed-loop stability 15% while clearly documenting architecture tradeoffs and experimental findings for senior researchers at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Wrote modular Java and Python components with 20+ unit and integration tests through Jenkins, PyTest, and Mockito, increasing nightly pipeline efficiency 68% across Linux, Windows, and macOS.
- Automated data logging, CSV reports, coverage summaries, and failure tracking while improving API response times 30% and collaborating with Microsoft and GitHub stakeholders during Agile delivery at scale.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Authored end-to-end architecture documentation, mapped 12+ integration points, and clearly presented operator-facing procedures to the client before implementing booking, payment, and real-time status workflows at production scale.
- Built user-friendly React and Flask interfaces across 50,000+ records, delivering sub-200-millisecond APIs and reliably reducing manual refund handling 70% through validated, production-ready customer booking workflows at scale.

Skills

Languages: C++, Python, Java, SQL, Shell, Bash, Linux, JavaScript
Autonomy/CV: Computer vision, PyTorch, trajectory data, HD maps, simulation, model validation, feature engineering
Pipelines: Logging, cloud storage, S3, AWS, data validation, reconciliation, ETL, monitoring, state tracking
Quality: Git, CI/CD, Jenkins, PyTest, Mockito, unit testing, integration testing, documentation, Agile

Projects

CPU-scheduling Simulator Live | *C++, Modular Algorithms*
- Implemented six C++ scheduling algorithms with edge-case handling, automated metrics, and operator-facing Gantt visualization, testing 1,000+ process combinations while reliably reducing average turnaround time 25% at scale.

CaptionCraft - Image Caption Generator Live | *Computer Vision, Python*
- Benchmarked BLIP, GIT, ViT-GPT-2, and custom vision architectures across 10,000 images and 100,000 captions, validating quality and improving measured accuracy from 81% to 96% through comparative analysis.

PiXMod Live | *Vision Models at Scale*
- Built a vision platform integrating CLIP, SAM, Stable Diffusion, and MongoDB, processing 100,000+ image embeddings while delivering semantic search, generative editing, scalable storage, and reliable cloud inference workflows.